# 2 Class and Object Namespace




## KEY IDEAS

- The class and **every instance** each own a namespace (a dict): `Class.__dict__` and `obj.__dict__` (same as `vars(obj)`).
- Lookup for `obj.x`: **1)** `obj.__dict__` then **2)** `type(obj).__dict__` then **3)** base classes in MRO order, else `AttributeError`.
- `obj.x = v` writes only to the **instance** namespace; changing `Class.x` is seen by every instance that has no own `x`.

**PITFALL** Mutable class attributes (`tags = []`) are **shared** by every instance. Create per-object lists/dicts inside `__init__`.

## EXAMPLE

```
class Counter:
    total = 0                   # in Counter.__dict__

    def __init__(self, label):
        self.label = label      # in obj.__dict__

a, b = Counter("a"), Counter("b")
print(a.__dict__)               # → {'label': 'a'}
print("total" in Counter.__dict__)  # → True
print("total" in a.__dict__)    # → False
print(a.total, b.total)         # → 0 0  (found in class)
Counter.total = 5               # change the class namespace
print(a.total, b.total)         # → 5 5  (shared)
```